

Predictive Analysis: Finding Body Fat

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Introduction

The purpose of this research and analysis is to build a regression model capable of predicting a gym member's body fat percentage. As a team, we completed detailed analysis using regression analysis tools to build a final, efficient model to predict the body fat percentage of a member at a local gym by measuring their physical characteristics. Using physical characteristics data acquired by our boss from a sample of 50 current members of the club, we developed a final model to perform this prediction. These characteristics are weight, height, age, neck size, chest size, abdomen size, hip size and thigh size. This report will explore and explain the steps taken to arrive at the final model, and apply the final model to solve for the predicted body fat percentage of a gym member, given that he is 36 years of age, weighs 185 pounds and is 72 inches tall. Other known measurements include his neck, chest, abdomen, hip, and thigh sizes which are 38.1, 102.0, 91.2, 102.6, 57.9 cm respectively. By building and applying a final regression model, this report aims to accurately solve for the gym member's predicted body fat percentage.

Conducting the Logic Step

In conducting the logic step, we constructed bivariate linear regression models and scatter plots to determine if there is a priori reason to expect a causal relationship between the predictors (age, weight, height, neck size, chest size, abdomen size, hip size, thigh size), and the response variable (percentage of body fat). To determine if there is a priori reason, three key statistics were examined:

1. Coefficient of the predictor
2. Coefficient of Determination (R^2)

3. ANOVA Analysis (F-Statistic and Significance F)

Body Fat Percentage and Age

Based on the statistics, we can determine that there is a priori reason to expect a causal relationship between age and body fat percentage. The coefficient of 0.559 tells us that as age increases, the response variable, body fat percentage, increases which means there is a positive relationship. For every 1 increase in age, the percentage of body fat increases by 0.559. The Coefficient of Determination (R^2) explains how much of the variation in percentage of body fat is explained by age. The R^2 of this model, 0.267682643, can be interpreted to mean that 28.6% of the variation in the response variable is explained by age. Therefore, suggesting that other variables may also influence the response. For the ANOVA analysis, we examined the F-statistic and the significance-F values to understand whether or not the regression model is statistically significant. The F-statistic is 17.545, which is the ratio of explained variation to unexplained variation in the model. Since the F-statistic is small, it tells us that there is more noise than signal in this model. The significance F value tells us the regression model is statistically significant because the significant F value, 0.000119341, is less than the $\alpha = 0.05$.

Body Fat Percentage and Weight

Based on the statistics, we can determine that there is a priori reason to expect a causal relationship between weight and body fat percentage. The coefficient of 0.138 tells us that as weight increases, the response variable, body fat percentage, increases which means there is a positive relationship. For every 1 increase in weight, the percentage of body fat increases by 0.138. The Coefficient of Determination (R^2) explains how much of the variation in percentage of body fat is explained by weight. The R^2 of this model, 0.374197271, can be interpreted to

mean that 37.4% of the variation in the response variable is explained by weight. Therefore, suggesting that other variables may also influence the response. The F-statistic is 28.701, which is the ratio of explained variation to unexplained variation in the model. The F-statistic is small but larger than the F-statistic for age, which tells us that this predictor may have more signal than noise. The significance F value tells us the regression model is statistically significant because the significant F value, 2.36316E-06, is less than the $\alpha = 0.05$.

Body Fat Percentage and Height

Based on the statistics, we can determine that there isn't a priori reason to expect a causal relationship between height and body fat percentage. The coefficient of -0.382 tells us that as height increases, the response variable, body fat percentage, decreases which means there is a negative relationship. For every 1 increase in height, the percentage of body fat decreases by -0.382. The Coefficient of Determination (R^2) explains how much of the variation in percentage of body fat is explained by height. The R^2 of this model, 0.070971242, can be interpreted to mean that 7.10% of the variation in the response variable is explained by height. Therefore, suggesting that height may explain a small amount of the percentage body fat. The F-statistic 3.666861319 is smaller than the F-statistic for age and weight, which tells us that this predictor may be less significant. The significance F value tells us the regression model is not statistically significant because the significant F value, 0.061474177, is greater than the $\alpha = 0.05$, which means that the predictor does not explain a significant portion of the variation in the response variable.

Body Fat Percentage and Neck Size

Based on the statistics, we can determine that there is a priori reason to expect a causal relationship between neck size and body fat percentage. The coefficient of 1.324 tells us that as the neck size increases, the response variable, body fat percentage, increases which means there is a positive relationship. For every 1 increase in neck size, the percentage of body fat decreases by 1.324. The Coefficient of Determination (R^2) explains how much of the variation in percentage of body fat is explained by the neck size. The R^2 of this model, 0.229658746, can be interpreted to mean that 23.0% of the variation in the response variable is explained by neck size. Therefore, suggesting that predictor may explain a certain amount of the percentage body fat. The F-statistic 14.31004733 is small which tells us that this predictor may be less significant. The significance F value tells us the regression model is statistically significant because the significant F value, 0.000429955, is less than the $\alpha = 0.05$.

Body Fat Percentage and Chest Size

Based on the statistics, we can determine that there is a priori reason to expect a causal relationship between chest size and body fat percentage. The coefficient of 0.618 tells us that as the chest size increases, the response variable, body fat percentage, increases which means there is a positive relationship. For every 1 increase in chest size, the percentage of body fat increases by 0.618. The Coefficient of Determination (R^2) explains how much of the variation in percentage of body fat is explained by the chest size. The R^2 of this model, 0.522038608, can be interpreted to mean that 52.2% of the variation in the response variable is explained by chest size. Therefore, suggesting that this predictor may significantly influence the response variable. For the ANOVA analysis, the F-statistic is 52.426. Since the F-statistic is comparatively large as compared to the previous predictors, but still small, it tells us that there may be more noise than

signal in this model for the predictor. The significance F value tells us the regression model is statistically significant because the significant F value, 3.14777E-09, is less than the $\alpha = 0.05$.

Body Fat Percentage and Abdomen Size

Based on the statistics, we can determine that there is a priori reason to expect a causal relationship between abdomen size and body fat percentage. The coefficient of 0.516 tells us that as the abdomen size increases, the response variable, body fat percentage, increases which means there is a positive relationship. For every 1 increase in abdomen size, the percentage of body fat increases by 0.516. The Coefficient of Determination (R^2) explains how much of the variation in percentage of body fat is explained by the abdomen size. The R^2 of this model, 0.6784753, can be interpreted to mean that 67.8% of the variation in the response variable is explained by abdomen size. Therefore, suggesting that this predictor may significantly influence the response variable. For the ANOVA analysis, the F-statistic is 101.2886862. Since the F-statistic is large, it tells us that there may be more signal than noise in this model for the predictor. The significance F value tells us the regression model is statistically significant because the significant F value, 2.05309E-13, is less than the $\alpha = 0.05$.

Body Fat Percentage and Hip

Based on the statistics, we can determine that there is a priori reason to expect a causal relationship between hip size and body fat percentage. The coefficient of 0.601 tells us that as the hip size increases, the response variable, body fat percentage, increases which means there is a positive relationship. For every 1 increase in the hip size, the percentage of body fat increases by 0.601. The Coefficient of Determination (R^2) explains how much of the variation in percentage of body fat is explained by the hip size. The R^2 of this model, 0.47972563, can be interpreted to mean that 48.0% of the variation in the response variable is explained by hip size. Therefore,

suggesting that this predictor may significantly influence the response variable since it can explain almost 50% of the variation. For the ANOVA analysis, the F-statistic is 44.25901325. Since the F-statistic is small, it tells us that there may be more noise than signal in this model for that predictor. The significance F value tells us the regression model is statistically significant because the significant F value, 2.50706E-08, is less than the $\alpha = 0.05$.

Body Fat Percentage and Thigh Size

Based on the statistics, we can determine that there is a priori reason to expect a causal relationship between thigh size and body fat percentage. The coefficient of 0.922 tells us that as the thigh size increases, the response variable, body fat percentage, increases which means there is a positive relationship. For every 1 increase in the thigh size, the percentage of body fat increases by 0.922. The Coefficient of Determination (R^2) explains how much of the variation in percentage of body fat is explained by the thigh size. The R^2 of this model, 0.465429624, can be interpreted to mean that 46.5% of the variation in the response variable is explained by the thigh size. Therefore, suggesting that this predictor may significantly influence the response variable. For the ANOVA analysis, the F-statistic is 41.79173212. Since the F-statistic is small, it tells us that there may be more noise than signal in this model for that predictor. The significance F value tells us the regression model is statistically significant because the significant F value, 4.87269E-08, is less than the $\alpha = 0.05$.

Conducting the FIT Step

To perform the FIT step, we constructed a multivariate regression model to predict body fat percentage with all predictors included. Using the ANOVA analysis tool in Excel, we derived a regression model for all of the treatments affecting body fat percentage. The results from this regression model are presented below in Table 1. Before interpreting the F-test, the model fit statistics provide important context. The regression output shows an R-Square value of 0.8412, which indicates that approximately 84% of the variation in body fat percentage is explained by all the predictors included in the model.

Table 1

Multiple Regression Summary Output for Predictors affecting Body Fat Percentage

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.917146334							
R Square	0.841157399							
Adjusted R Square	0.81016372							
Standard Error	3.956742199							
Observations	50							
ANOVA								
	df	SS	MS	F	Significance F			
Regression	8	3399.144638	424.8930797	27.1396441	4.83E-14			
Residual	41	641.888162	15.65580883					
Total	49	4041.0328						
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95%	Upper 95%
Intercept	-35.43089037	24.90402585	-1.422697301	0.16238835	-85.72559035	14.86380961	-85.72559035	14.86380961
Age	0.090499491	0.088013055	1.028250761	0.309860109	-0.087246478	0.268245461	-0.087246478	0.268245461
Weight	-0.192808347	0.078315214	-2.461952625	0.018112479	-0.350969128	-0.034647565	-0.350969128	-0.034647565
Height	-0.064226408	0.115960259	-0.553865684	0.582679094	-0.2984129	0.169960084	-0.2984129	0.169960084
Neck	-0.334840695	0.402273545	-0.832370658	0.410022058	-1.147248592	0.477567201	-1.147248592	0.477567201
Chest	0.023860488	0.178815717	0.133436189	0.894501249	-0.337265174	0.38498615	-0.337265174	0.38498615
Abdomen	0.913237371	0.16396954	5.569554979	1.76849E-06	0.582094169	1.244380572	0.582094169	1.244380572
Hip	-0.310677796	0.274879004	-1.130234725	0.264946084	-0.8658072	0.244451609	-0.8658072	0.244451609
Thigh	0.778651907	0.29072781	2.678284912	0.010598442	0.19151519	1.365788624	0.19151519	1.365788624

The results from the table provide us with an important finding: the F-statistic, which is highlighted for reference. To find the overall significance of the model, we need to compare the F-statistic and the F-critical values. Using the values from the table, the F-critical value can be

solved for. To calculate the F-critical value, with a 0.05 significance level, two values are needed: the first degree of freedom (K) or the number of predictors, and the second degree of freedom.

Using the ANOVA table, these values have been provided for us, where the first degree of freedom is 8 and the second degree of freedom is 41. Using the excel function:

$F.INV.RT(0.05,8,41)$, the F-critical value is calculated, and when rounded to 2 decimal places, is approximately 2.17.

By comparing the F-critical to the F-statistic, an F-test was performed to determine if the overall regression model suggests a significant relationship between all included predictors and the response variable. Rounded to two decimal places, the F-critical value is 2.17, whereas the F-statistic value is 27.14. Comparing these values, we observed that the F-statistic value is greater than the F-critical value. Our null hypothesis states that all predictors are equal to zero, where, $H_0: \mu_1 = \mu_2 = \mu_3 = \dots = 0$. On the other hand, our alternative hypothesis asserts that at least one predictor is different to zero, where, H_A : At least one $\mu_i \neq 0$. Since the F-statistic value is greater than the F-critical value, we reject our null hypothesis. By rejecting the null hypothesis, our analysis asserts that at least one of the treatments used in the model is significant. This indicates that at least one of the group means is different from the other treatment groups. It is important to note that a significant overall F-test does not imply that all predictors are individually significant. Instead, it confirms that the set of predictors collectively improves the model. This result justifies moving forward to examine the individual significance values for each predictor to determine which predictors significantly influence body fat percentage at the 0.05 significance level.

Model Refinement

Parsimony

Having analyzed the significance of all predictors in the overall model in the previous step, we will now determine if each predictor meaningful contributes to explaining the body fat percentage. By comparing the p-value of each predictor to the alpha level of 0.05 (5.00%), we determined which treatments were perhaps significant. Predictors with p-values less than the alpha level of 0.05 were considered significant to our model; whereas, p-values greater than the alpha were not considered significant. Using the p-values from Table 1, we used the IF function in Excel to compare the p-values to the alpha level and to classify each predictor as perhaps significant or perhaps not significant. The results are presented below in Table 2.

Table 2

Summary of Significance Testing of Individual Regression Predictors at $\alpha = 0.05$ using P-Values

Alpha		5.00%	
	P-value	Less than 0.05	Interpretation
Intercept	0.16238835	N	Perhaps Not Significant
Age	0.309860109	N	Perhaps Not Significant
Weight	0.018112479	Y	Perhaps Significant
Height	0.582679094	N	Perhaps Not Significant
Neck	0.410022058	N	Perhaps Not Significant
Chest	0.894501249	N	Perhaps Not Significant
Abdomen	1.76849E-06	Y	Perhaps Significant
Hip	0.264946084	N	Perhaps Not Significant
Thigh	0.010598442	Y	Perhaps Significant

The results shown above highlight three predictors: weight, abdomen and thigh as being less than 0.05, which indicates that these predictors are most likely significant to our model. In contrast, the predictors such as age, height, neck, chest and hip were greater than 0.05 and

therefore, we considered them to be insignificant to our model. By removing predictors that are not statistically significant, we create a more efficient and accurate model without sacrificing predictive power. Based on the results from the significance testing, it may be appropriate to exclude age, height, neck, chest, and hip from the final model. These variables do not appear to contribute meaning to the body fat percentage, compared to weight, abdomen, and thigh. Therefore, moving forward, we expect weight, abdomen, and thigh to serve as the core predictors in our regression model. The implication of this decision is that our refined model will be simpler, easier to interpret and more efficient as we only expect three of the eight treatments to be included in the final model. Thus we will consider and expect the predictors such as weight, abdomen and thigh to be the only significant predictors in determining body fat percentage.

Stability

To evaluate the stability of our regression model, we examined the correlation matrix for all predictors. The purpose of this step is to identify whether any predictors are strongly correlated with each other, which can introduce multicollinearity. Multicollinearity inflates standard errors, weakens t-tests, which makes it difficult to determine which predictors are truly significant to the model. Table 3, as shown below, displays the correlation matrix for all predictors. The highlighted cells have correlation values greater than or equal to 0.80, indicating a moderately strong to extremely strong correlation between two predictors.

Table 3

Correlation Matrix

	Fat%	Age	Weight	Height	Neck	Chest	Abdomen	Hip	Thigh
Fat%	1								
Age	0.517381	1							
Weight	0.611717	0.26524	1						
Height	-0.2664	-0.27631	0.108979	1					
Neck	0.479227	0.176075	0.882158	0.200786	1				
Chest	0.722522	0.375946	0.912335	0.013556	0.820394	1			
Abdomen	0.823696	0.442304	0.914592	-0.05248	0.781216	0.942089	1		
Hip	0.692622	0.313798	0.959475	-0.04544	0.803797	0.911319	0.942021	1	
Thigh	0.682224	0.218528	0.937201	-0.03657	0.822721	0.859122	0.8903	0.937727	1

Looking at highlighted values in the correlation matrix, we observed several extremely high correlations among the predictors. The predictors weight, chest, abdomen, hip, and thigh are all strongly correlated with each other, with many correlations exceeding 0.80. Whereas, age and height have weaker correlations. In particular, we see that weight, chest, abdomen, hip, and thigh all show correlations above 0.90 with one another. Neck is also moderately correlated with these variables with correlations ranging from 0.78 to 0.82. Abdomen is strongly correlated with most predictors with values as high as 0.94. This indicates substantial multicollinearity. Since predictors are strongly correlated, these variables provide overlapping information, which can inflate standard errors and affect the stability of t-tests in the regression model. Hence, including all predictors in the regression model introduces multicollinearity and makes it difficult to determine which predictors are truly significant to the model.

When predictors are this strongly correlated, they compete to explain the same portion of variance in the response variable. As a result, their individual t-tests may appear non-significant even if the variables are significant. This helps explain why in the full model, only weight, abdomen, and thigh emerged as significant predictors. These variables likely capture the majority of the shared variance among the highly correlated predictors. In contrast, predictors such as chest, hip, and neck may have appeared non-significant because their effects are absorbed by stronger, highly correlated predictors.

Given this level of multicollinearity, including multiple predictors that are redundant adds noise. Therefore, removing non-significant predictors such as chest, hip, neck, height, and age helps create a more stable and efficient model. Overall, the correlation matrix shows substantial multicollinearity among the treatments. This justifies excluding several predictors from the final

model and retaining only the variables that remain significant, such as weight, abdomen, and thigh. This decision improves the stability and parsimony of the regression model.

Performing Forward Stepwise Regression

Forward Stepwise Regression helps us find the best set of predictor variables for explaining a dependent variable, where the dependent variable is body fat percentage, and the possible predictors are age, weight, height, neck, chest, abdomen, hip, and thigh.

In our first step, we tested each possible predictor individually against the body fat percentage values. This led to eight different ANOVA tables, which provided us with the p-values, which were used to compare the predictors' significance based on our confidence level being 95%. The abdomen predictor had the smallest p-value of 0.000000000000205 making it the first predictor to be included in our model. In the second step, we tested the remaining predictors against body fat percentage and abdomen as our model. After running each test and comparing the p-values in the seven ANOVA tables, the next predictor to have the smallest p-value of 0.0000022675 was weight. In the third step, we proceeded to repeat this process and tested the remaining predictors against our new model, which included abdomen and weight as significant variables. After testing each remaining predictor, the next significant predictor found in the remaining ANOVA tables was thigh with a P-value of 0.010463323.

In our fourth step, none of the remaining predictors had a small enough p-value to be considered significant variables for body fat percentage. All of the remaining p-values (age = 0.178527324494291, height size = 0.510528307603222, neck size = 0.498561474042276, chest size = 0.957891409951828, and hip = 0.429450121264071, respectively) were all greater than our alpha value of 0.05, making them insignificant. Therefore, we concluded that no additional

predictors should be added to our model. Additionally, this aligned with our predicted significant variables that we observed in the parsimony step, as we hypothesized that abdomen, thigh, and weight would be the only significant predictors in our model.

Performing Backwards Stepwise Regression

To ensure the accuracy of our regression model, we performed a backwards stepwise regression. With backward regression, we looked at the p-value to figure out which variable is insignificant to our model. The predictor with the highest p-value was eliminated, and the regression model was subsequently re-run. We also analyzed the Adjusted R-Square as a second metric, which determined whether the removal of a predictor created a better model. During the first step, chest size had the highest p-value of .8945; hence, we eliminated chest size as a predictor from our model. In the next step, height was eliminated with a p-value of .572424588. In the following step, neck size was eliminated with a p-value of .377652562. After this, hip size was eliminated with a p-value of .442571716.

Lastly, we eliminated age, with a p-value of .17852732. When we removed chest, height, neck and hip, the Adjusted R-square value steadily increased every time, signifying that the eliminated variables did not contribute meaningful explanatory power to the model and the removal improved the overall model by reducing unnecessary complexity. In the last step, in which we eliminated age, we saw a decrease in the Adjusted R-Square from .819983523 to .816587824. This suggests that despite its p-value exceeding the 0.05 threshold, the removal of age contributed meaningfully to the model when considered alongside other predictors. After this process, we are left with weight, abdomen size and thigh size as significant predictors to our

model. Overall, the results of performing the backwards stepwise regression were similar to those of the forward stepwise regression.

Model Improvement

To improve the model, we decided to transform the data provided for age into binary data. We transformed the data by setting any value greater than 30, equal to a value of '1' and setting any value less than or equal to 30, equal to a value of '0'. The following tables below illustrate the results of the final model before and after binary encoding.

Table 4

Results showing Adjusted R^2 and Weight, Abdomen, Thigh Coefficients before Binary Encoding

Regression Statistics				
Multiple R	0.90984457			
R Square	0.827817141			
Adjusted R Square	0.816587824			
Standard Error	3.889217481			
Observations	50			

	Coefficients	Standard Error	t Stat	P-value
Intercept	-59.00212117	8.299337126	-7.109257074	6.24951E-09
Weight	-0.277079512	0.045842382	-6.044177935	2.48614E-07
Abdomen	0.95026655	0.097426599	9.753666456	8.94285E-13
Thigh	0.651426112	0.244032203	2.669426837	0.010463323

Table 5

Results showing Adjusted R^2 and Weight, Abdomen, Thigh Coefficients after Binary Encoding

Regression Statistics				
Multiple R	0.90984457			
R Square	0.827817141			
Adjusted R Square	0.816587824			
Standard Error	3.889217481			
Observations	50			

	Coefficients	Standard Error	t Stat	P-value
Intercept	-59.00212117	8.299337126	-7.109257074	6.25E-09
Weight	-0.277079512	0.045842382	-6.044177935	2.49E-07
Abdomen	0.95026655	0.097426599	9.753666453	8.94E-13
Thigh	0.651426112	0.244032203	2.669426839	0.010463323

In comparing the values of the coefficients of the three significant predictors: weight, abdomen and thigh, before and after the binary encoding, we can observe that the values do not change. As seen in Table 4 and Table 5, the coefficients for weight, abdomen and thigh are as follows respectively: -0.277079512, 0.95026655 and 0.651426112. The same observation can be made for the adjusted R^2 value. As seen in Table 4 and Table 5, the adjusted R^2 value before and after binary encoding is 0.816587824. Therefore, we can conclude that binary encoding the data for age does not significantly improve or change the final prediction model. We can assume that our original model before improvement is as accurate as the model after transformation. We decided to build our final prediction model based on these results and findings.

Implementation and Conclusion

Solution to Manager's Problem

The manager's objective was to estimate the body fat percentage of a gym member using the regression model developed in the project. After conducting detailed analysis and review, our final Regression model is: $\hat{y} = -59.002 + 0.950X_1 + -0.277X_2 + 0.651X_3$; where, X_1 is abdomen size, X_2 is weight and X_3 is thigh size. Here, the co-efficient value for each predictor has been rounded to three decimal places for simplicity, but the unrounded co-efficient values were used in solving for the body fat percentage of the gym member. Using the final refined model with the predictors weight, abdomen, and thigh, the body fat percentage was calculated by substituting the given values (abdomen = 91.2, weight = 185, thigh = 57.9) into the regression equation. The predicted body fat percentage using our final regression model is 14.1%.

This result shows how the final model can be directly applied in a real-world setting. Instead of needing all body measurements, the gym can efficiently estimate body fat using only

the most significant variables identified in the analysis. The prediction is also consistent with the model's strong performance, since these variables were proven to be statistically significant and provided high explanatory power. Therefore, the manager can confidently use this regression model as a practical tool for estimating body fat percentage for clients with similar characteristics.

Summary

This project developed a multiple regression model to predict body fat percentage using body measurements. While the initial model included many variables, it suffered from multicollinearity, as many predictors were highly correlated (often above 0.80), which reduced the model's stability and made some variables appear statistically insignificant. To address this, the model was refined to focus only on the most impactful predictors. The final regression model includes weight, abdomen, and thigh, the only variables with statistically significant p-values, which were all less than the significance level of 0.05. Among these, abdomen size is the strongest predictor, demonstrating the greatest influence on body fat percentage. This indicates that increases in abdomen measurements are strongly associated with increases in body fat, making it the most reliable single indicator.

By simplifying the model and removing overlapping variables, the final model achieved both stability and high predictive power, explaining approximately 83% of the variation in body fat ($R^2 \approx 0.83$). The model was also successfully applied to the manager's problem, producing a predicted body fat percentage of 14.1%, demonstrating its practical usefulness. Although effective, the model has some limitations, including a relatively small sample size and potential lack of generalizability to other populations. Future improvements could involve collecting more data and exploring additional variables to further enhance accuracy.

Overall, this project shows that body fat percentage can be accurately and efficiently estimated using a small set of key measurements, making the model both statistically reliable and valuable for real-world applications.